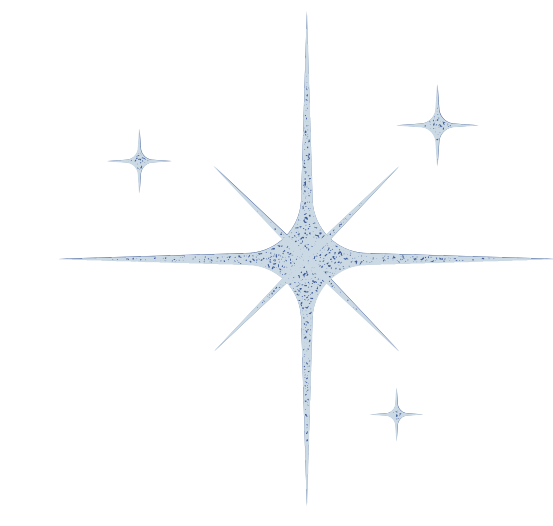


Analyzing VLTI/GRAVITY Data of Beta Pictoris b for Exomoon-Induced Wobble

Saji Alqadhi, Emma Zheng, Madeleine Morse-Hill, Nhi Nguyen, and Arbar Sharif

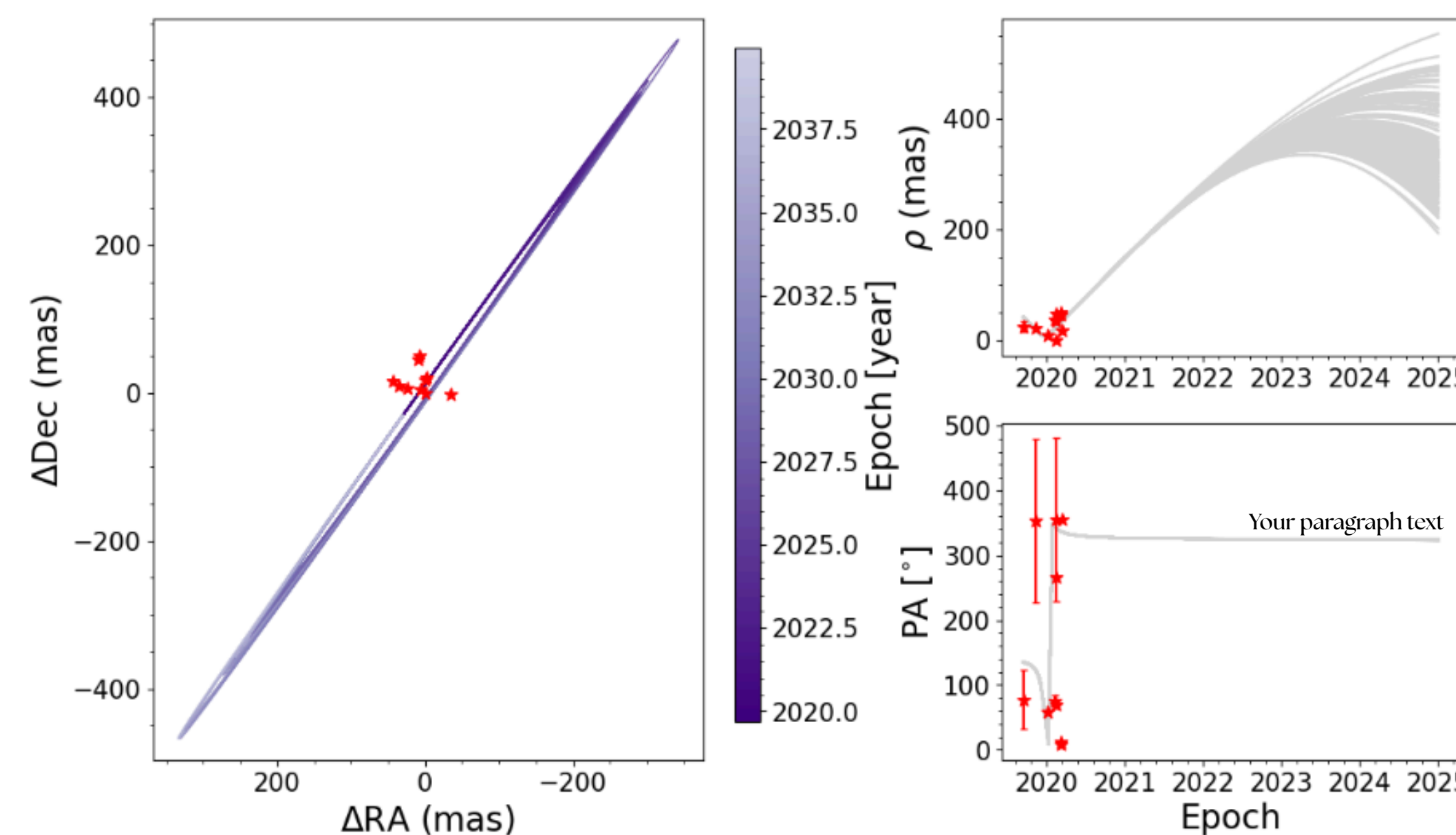


Background

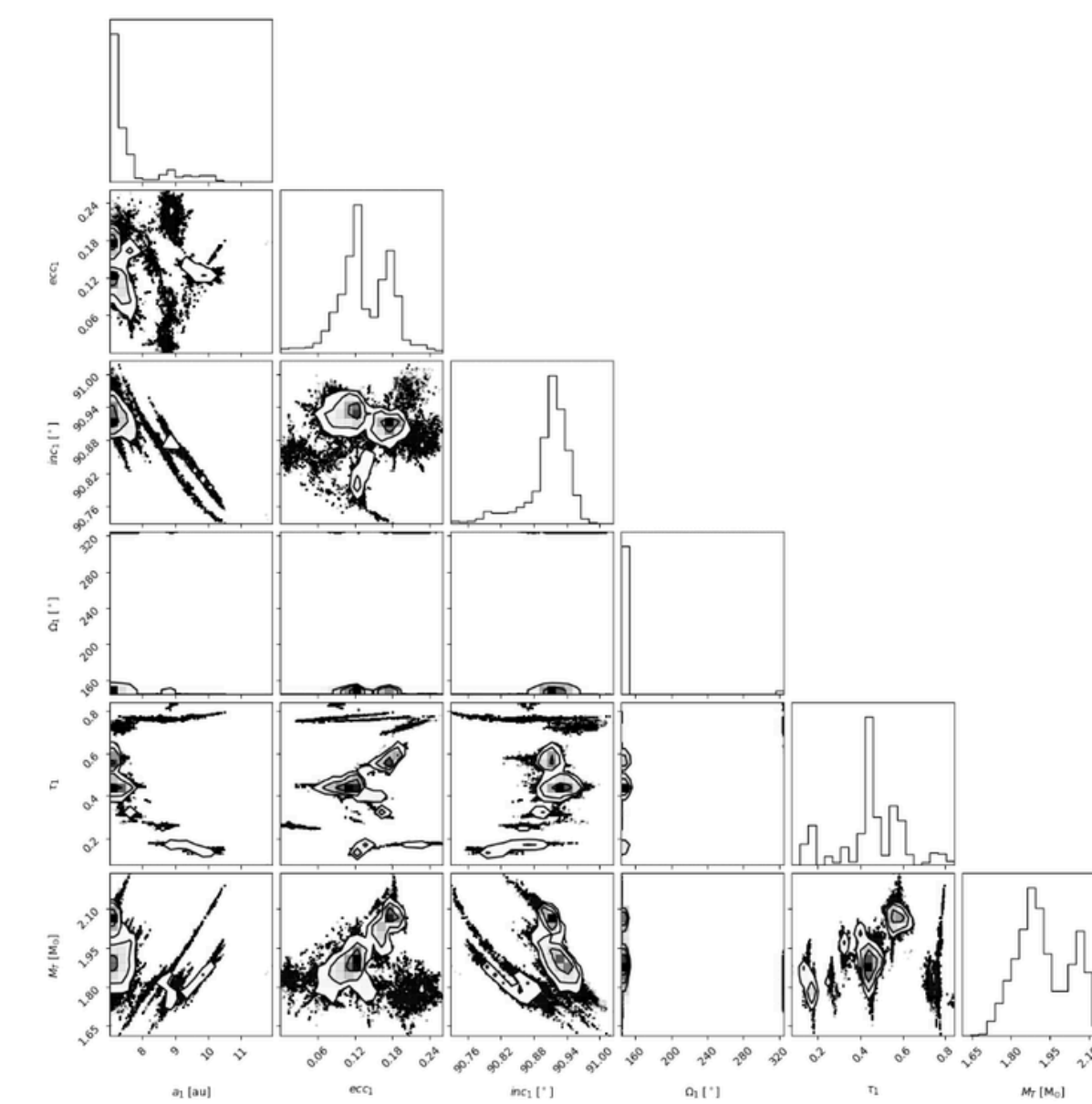
- Exomoons: moons that orbit outside our solar system
 - Small gravitational effects compared to planets
 - None confirmed
- Beta Pic b: Best planet to analyze orbiting exomoon in the short term (Kral et. al 2025)
 - Super-Jupiter ($\sim 12M_J$)
 - A-type star Beta Pictoris
 - Orbital radius: ~ 10 AU
 - 64 light years away
 - Young, active planetary system

Main Findings

We did not conclusively detect an exomoon around Beta Pic b based on its motion due to lack of data precision and accuracy, as well as many potentially confounding variables associated with young planetary systems.



Simulated Keplerian orbit from *orbitize!* for Beta Pic b including graph of angular separation between the planet and the star (ρ) and position angle (PA).



Method

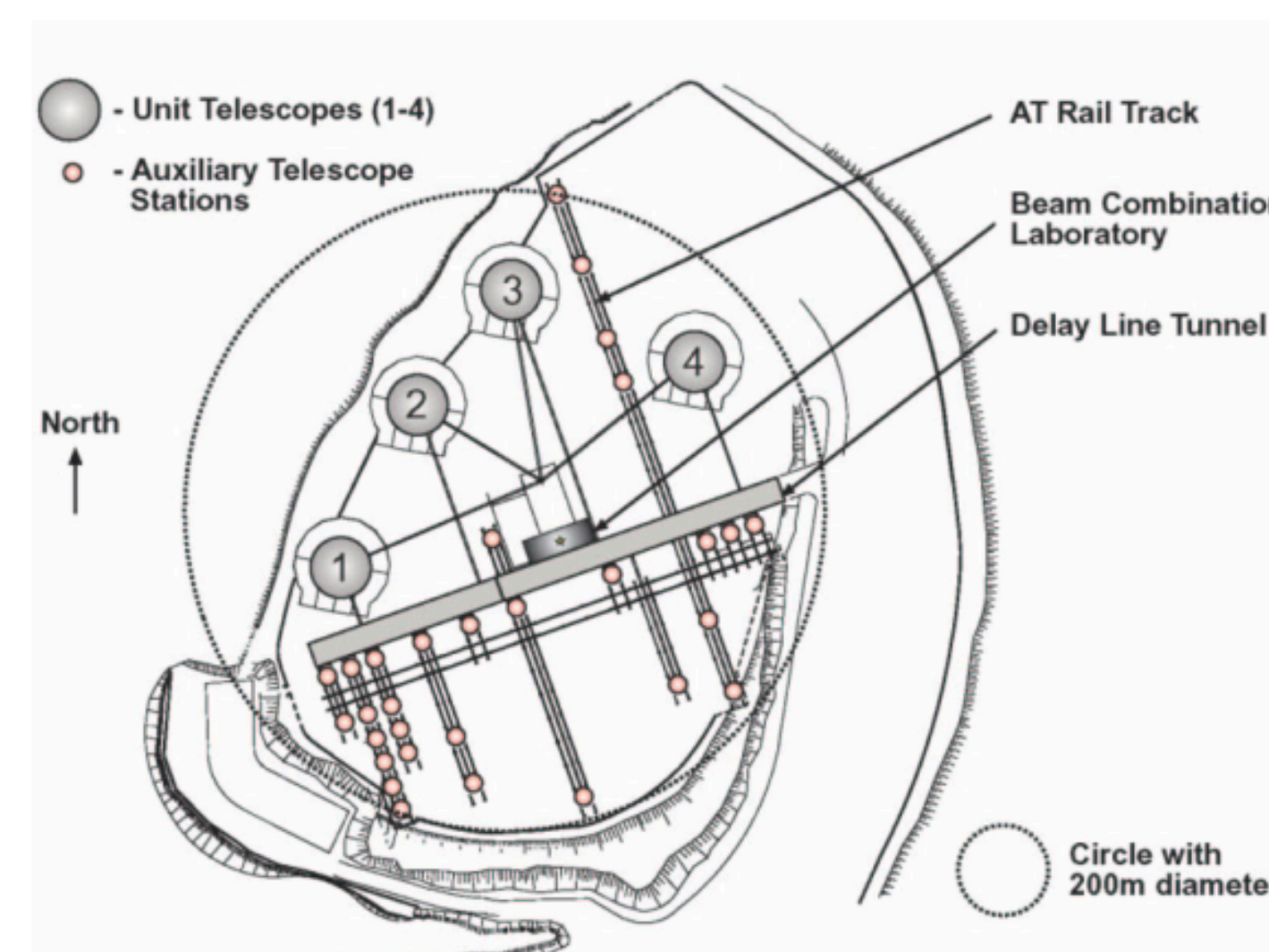
We modelled the motion of Beta Pic b using 9 epochs of data from VLTI/GRAVITY, then analyzed it for a wobble that would signify the presence of an exomoon.

Our Steps:

- Make data usable: process using *EsoRex* and GRAVITY Pipeline
- Refine data: reduce noise and stellar data using *ExoGravity* and convert to RA/Dec offsets
- Run simulations of orbit on *orbitize!* Fitting then subtracting normal Keplerian orbit
- Analyze residuals for wobble whether noise or a moon explains the wobble better

VLTI/GRAVITY

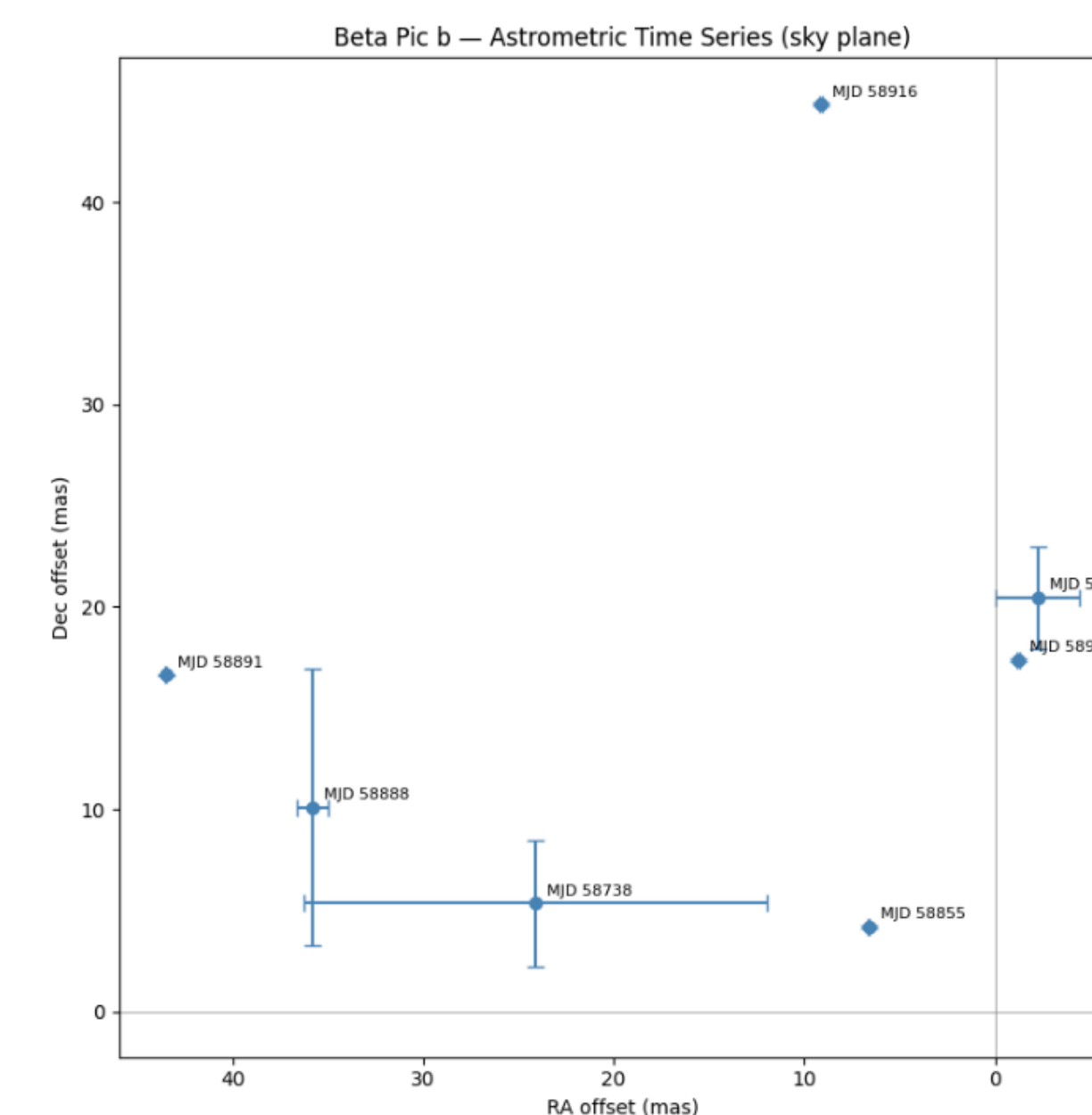
- Very Large Telescope Interferometer: infrared interferometry (K-band from 2.0 to $2.4 \mu\text{m}$) reduces stellar glare and dust
- Directly measures exoplanet instead of transits!
- 4 telescope (8.2 m diameter each) beams combine for high precision (50 microarcseconds),



Final Very Large Telescope Interferometer Layout from VLTI Tutorial: VLTI Concept & technical aspects, B. Koehler, 20 November 2001, Garching



Link to our presentation!



RA/Dec offsets of 8 of our epochs including their errors graphed (9th was cut out due to extremely large error)

Epoch	RA Offset	Dec Offset
2019-09-12	0.0	0.0
2019-11-09	146.45959	250.35176
2019-11-11	0.0	0.0
2020-01-07	-0.00096	-0.0025
2020-02-09	-72.8173	-43.0875
2020-02-10	-69.81	-116.728
2020-02-12	-67.31	-112.728
2020-03-08	-72.072	-118.118
2020-03-12	-72.072	-118.118

RA/Dec offsets of all of our epochs including their errors graphed after cleaning and reduction